

COMPREHENSIVE PRACTICE PROBLEMS & SOLUTIONS

Relations and Functions - DMST Module

SECTION A: CARTESIAN PRODUCT PROBLEMS

Level 1: Basic (Warm-up)

Problem A1.1: Find $A \times B$ if $A = \{1, 2\}$, $B = \{a, b, c\}$.

Solution:

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$$

Cardinality: $|A \times B| = 2 \times 3 = 6 \checkmark$

Problem A1.2: If $|A| = 4$ and $|B| = 5$, find $|A \times B|$ and $|B \times A|$.

Solution:

- $|A \times B| = 4 \times 5 = 20$
- $|B \times A| = 5 \times 4 = 20$
- (Note: Cardinality is the same, but elements are different: $(a,b) \neq (b,a)$)

Problem A1.3: Is it true that $(2, 3) = (3, 2)$? Explain.

Solution: No, ordered pairs are not sets. $(2, 3) \neq (3, 2)$ because:

- In $(2, 3)$: first element is 2, second is 3
- In $(3, 2)$: first element is 3, second is 2
- The positions matter in ordered pairs

Level 2: Intermediate

Problem A2.1: If $A = \{x, y, z\}$, $B = \{3, 4\}$, find $B \times A$ and compare with $A \times B$.

Solution:

- $A \times B =$
- $B \times A =$
- **Comparison:** Different sets because order of sets matters; $A \times B$ contains ordered pairs starting with set A elements, while $B \times A$ contains ordered pairs starting with set B elements
- Both have $|A \times B| = |B \times A| = 6$ elements, but they're NOT equal

Problem A2.2: Find X and Y if $X \times Y = \{(a, 5), (a, 6), (b, 5), (b, 6)\}$.

Solution:

- First coordinates of all pairs: $\{a, b\} \rightarrow \mathbf{X = \{a, b\}}$
- Second coordinates of all pairs: $\{5, 6\} \rightarrow \mathbf{Y = \{5, 6\}}$
- Verification: $|X| = 2$, $|Y| = 2$, $|X \times Y| = 2 \times 2 = 4 \checkmark$

Problem A2.3: Given $|A \times B| = 12$ and $|A| = 4$, find $|B|$.

Solution:

- Formula: $|A \times B| = |A| \times |B|$
- Therefore: $12 = 4 \times |B|$

- $|B| = 3$

Level 3: Advanced

Problem A3.1: If $A = \{1, 2\}$, $B = \{m, n\}$, $C = \{p, q, r\}$, find $|A \times B \times C|$.

Solution:

- $|A \times B \times C| = |A| \times |B| \times |C| = 2 \times 2 \times 3 = \mathbf{12}$
- Elements are ordered triples like $(1, m, p)$, $(1, m, q)$, etc.

Problem A3.2: Prove that if $A \times B = B \times A$, then $A = B$.

Solution:

Proof:

Assume $A \times B = B \times A$.

Suppose $(a, b) \in A \times B$ for some $a \in A$, $b \in B$.

Since $A \times B = B \times A$, we have $(a, b) \in B \times A$.

This means $a \in B$ and $b \in A$.

Therefore, every element of A is in B , so $\mathbf{A \subseteq B}$.

By similar reasoning, $\mathbf{B \subseteq A}$.

Therefore, $\mathbf{A = B}$. ✓

Problem A3.3: Verify $A \times (B \cap C) = (A \times B) \cap (A \times C)$ for $A = \{1, 2\}$, $B = \{a, b\}$, $C = \{b, c\}$.

Solution:

LHS: $A \times (B \cap C)$

- $B \cap C =$
- $A \times \{b\} = \{(1,b), (2,b)\} \dots \textcircled{1}$

RHS: $(A \times B) \cap (A \times C)$

- $A \times B =$
- $A \times C =$
- $(A \times B) \cap (A \times C) = \{(1,b), (2,b)\} \dots \textcircled{2}$

$\textcircled{1} = \textcircled{2} \checkmark$ **Verified**

SECTION B: RELATION PROBLEMS

Level 1: Basic Recognition

Problem B1.1: Let $A = \{1, 2, 3\}$ and $R = \{(1,2), (2,3)\}$. Is R a relation on A ?

Solution:

Yes, R is a relation on A because $R \subseteq A \times A$. $\checkmark$

(Note: It's a relation from A to A)

Problem B1.2: Find Domain and Range of $R = \{(2,4), (3,6), (4,8)\}$.

Solution:

- **Domain** =
- **Range** =

Problem B1.3: If $R = \{(1,2), (2,3), (3,1)\}$, find R^{-1} .

Solution:

$$R^{-1} = \{(2, 1), (3, 2), (1, 3)\}$$

Level 2: Property Verification

Problem B2.1: Let $A = \{1, 2, 3\}$ and $R = \{(1,1), (2,2), (3,3)\}$. Identify all properties of R .

Solution:

- **Reflexive:** Yes ✓ (all elements (a,a) are in R)
- **Symmetric:** Yes ✓ (R contains only (a,a) pairs, which are symmetric)
- **Transitive:** Yes ✓ (only (a,a) pairs, trivially transitive)
- **Antisymmetric:** Yes ✓ (if $(a,b), (b,a) \in R$, then $a = b$ always holds)
- **Equivalence:** Yes ✓ (reflexive, symmetric, transitive)

Conclusion: R is an **equivalence relation**

Problem B2.2: Let $A = \{1, 2, 3, 4\}$ and $R = "x \text{ divides } y"$. Determine properties.

Solution:

Setting up R :

$R =$

- **Reflexive:** Yes ✓ ($a|a$ for all a)
- **Symmetric:** No ✗ ($2|4$ but $4 \nmid 2$)
- **Transitive:** Yes ✓ (if $a|b$ and $b|c$, then $a|c$)
- **Antisymmetric:** Yes ✓ (if $a|b$ and $b|a$, then $a=b$)

Conclusion: R is a **partial order** (reflexive, transitive, antisymmetric)

Problem B2.3: Test if $R = \{(1,2), (2,1), (2,3), (3,2), (1,3), (3,1)\}$ is symmetric and transitive.

Solution:

Symmetric:

- $(1,2) \in R$ and $(2,1) \in R$ ✓
- $(2,3) \in R$ and $(3,2) \in R$ ✓
- $(1,3) \in R$ and $(3,1) \in R$ ✓
- **R is symmetric** ✓

Transitive:

- $(1,2) \in R$ and $(2,1) \in R$, but $(1,1) \notin R$ ✗
- **R is NOT transitive** ✗

Level 3: Equivalence Relations

Problem B3.1: Let $I = \{\text{all integers}\}$, and $R = \{(a,b) \mid a \equiv b \pmod{5}\}$. Prove R is an equivalence relation.

Solution:

Reflexive: $a \equiv a \pmod{5}$ because $a - a = 0 = 5 \cdot 0$, so $5 \mid (a-a)$ ✓

Symmetric: If $a \equiv b \pmod{5}$, then $5 \mid (a-b)$, so $a-b = 5k$.

Then $b-a = -5k = 5(-k)$, so $5 \mid (b-a)$, thus $b \equiv a \pmod{5}$ ✓

Transitive: If $a \equiv b \pmod{5}$ and $b \equiv c \pmod{5}$, then:

- $a - b = 5m$ and $b - c = 5n$
- $a - c = (a-b) + (b-c) = 5m + 5n = 5(m+n)$
- So $5 \mid (a-c)$, thus $a \equiv c \pmod{5}$ ✓

Conclusion: R is an equivalence relation ✓

Equivalence Classes:

- $[0] =$
- $[1] =$

- $[2] =$
- $[3] =$
- $[4] =$

Problem B3.2: Let $A = \{\text{lines in a plane}\}$, and $R = \text{"is parallel to"}$. Is R an equivalence relation?

Solution:

Reflexive: Is every line parallel to itself?

- If we define a line as parallel to itself (standard definition): Yes ✓
- If a line is not parallel to itself: No ✗

(Assuming standard convention that a line is parallel to itself)

Symmetric: If line $l_1 \parallel l_2$, then $l_2 \parallel l_1$? **Yes** ✓

Transitive: If $l_1 \parallel l_2$ and $l_2 \parallel l_3$, then $l_1 \parallel l_3$? **Yes** ✓

Conclusion: R is an **equivalence relation** ✓

SECTION C: FUNCTION PROBLEMS

Level 1: Basic Function Recognition

Problem C1.1: Is $R = \{(1,2), (2,4), (3,6)\}$ a function?

Solution:

- Domain = $\{1, 2, 3\}$ has 3 elements
- Each element maps to exactly one value
- No input maps to multiple outputs

- **Yes, R is a function** ✓

Problem C1.2: Is $R = \{(1,2), (1,3), (2,4)\}$ a function?

Solution:

- Element 1 maps to both 2 and 3
- Violates the definition (must be exactly one output per input)
- **No, R is NOT a function** ✕

Problem C1.3: For $f = \{(0,1), (1,3), (2,5), (3,7)\}$, find domain and range.

Solution:

- **Domain** =
- **Range** =

Level 2: Injectivity and Surjectivity

Problem C2.1: Is $f(x) = 2x + 1$ on $f: \mathbb{R} \rightarrow \mathbb{R}$ injective? Surjective? Bijective?

Solution:

Injective:

Assume $f(a) = f(b)$

$$\Rightarrow 2a + 1 = 2b + 1$$

$$\Rightarrow 2a = 2b$$

$$\Rightarrow a = b$$

Yes, injective ✓

Surjective:

For any $y \in \mathbb{R}$, solve $f(x) = y$

$$\Rightarrow 2x + 1 = y$$

$$\Rightarrow x = (y-1)/2 \in \mathbb{R}$$

Yes, surjective ✓

Bijjective: Yes ✓ (both injective and surjective)

Inverse Function:

From $x = (y-1)/2$, we get $f^{-1}(x) = (x-1)/2$

Problem C2.2: Is $f(x) = x^2$ on $f: \mathbb{R} \rightarrow \mathbb{R}$ injective? Surjective? Bijjective?

Solution:

Injective:

$f(2) = 4$ and $f(-2) = 4$, but $2 \neq -2$

Not injective ✕

Surjective:

Can we get $f(x) = -1$? We need $x^2 = -1$, which has no real solution

Not surjective ✕

Bijjective: No ✕

Modification: If we change codomain to $\mathbb{R}_{\geq 0}$, then $f: \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ becomes surjective. If we restrict domain to $\mathbb{R}_{\geq 0}$, then $f: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ becomes bijective.

Problem C2.3: Is $f: \{1,2,3\} \rightarrow \{a,b,c\}$ defined by $f = \{(1,a), (2,b), (3,c)\}$ bijective?

Solution:

- **Injective:** All different inputs map to different outputs ✓
- **Surjective:** Range = $\{a,b,c\}$ = Codomain ✓
- **Bijjective:** Yes ✓

Inverse:

$$f^{-1} =$$

Level 3: Function Composition

Problem C3.1: Given $f(x) = x + 2$ and $g(x) = 3x$, find $(f \circ g)(x)$ and $(g \circ f)(x)$.

Solution:

$(f \circ g)(x)$:

$$= f(g(x))$$

$$= f(3x)$$

$$= 3x + 2$$

$(g \circ f)(x)$:

$$= g(f(x))$$

$$= g(x + 2)$$

$$= 3(x + 2)$$

$$= 3x + 6$$

Observation: $(f \circ g)(x) \neq (g \circ f)(x)$, confirming composition is NOT commutative

Problem C3.2: If $f(x) = 2x - 1$ and $g(x) = (x+1)/2$, verify that $g = f^{-1}$.

Solution:

Method 1: Check $f(g(x)) = x$:

$$f(g(x)) = f((x+1)/2)$$

$$= 2 \cdot (x+1)/2 - 1$$

$$= (x+1) - 1$$

$$= x \quad \checkmark$$

Method 2: Check $g(f(x)) = x$:

$$\begin{aligned} g(f(x)) &= g(2x-1) \\ &= ((2x-1)+1)/2 \\ &= 2x/2 \\ &= x \quad \checkmark \end{aligned}$$

Conclusion: $g = f^{-1} \quad \checkmark$

Problem C3.3: Given $f(x) = x^2$ and $g(x) = \sqrt{x}$, find $(f \circ g)(x)$ and $(g \circ f)(x)$.

Solution:

$(f \circ g)(x)$:

$$\begin{aligned} &= f(g(x)) \\ &= f(\sqrt{x}) \\ &= (\sqrt{x})^2 \\ &= x \quad (\text{for } x \geq 0) \end{aligned}$$

$(g \circ f)(x)$:

$$\begin{aligned} &= g(f(x)) \\ &= g(x^2) \\ &= \sqrt{x^2} \\ &= |x| \end{aligned}$$

Note: These are NOT the same unless we restrict $x \geq 0$

SECTION D: PIGEON-HOLE PRINCIPLE

Level 1: Direct Application

Problem D1.1: 13 people, how many must share the same month of birth?

Solution:

- $n = 13$ (people)
- $m = 12$ (months)
- Minimum = $\lceil 13/12 \rceil = \lceil 1.08 \rceil = \mathbf{2 \text{ people}}$

Problem D1.2: 100 students, 26 characters in alphabet (A-Z), how many same letter?

Solution:

- $n = 100$ (students)
- $m = 26$ (letters)
- Minimum = $\lceil 100/26 \rceil = \lceil 3.85 \rceil = \mathbf{4 \text{ students}}$ with same starting letter

Level 2: Reverse Application

Problem D2.1: To guarantee 5 people share the same favorite color from k colors, how many people needed?

Solution:

- We want $\lceil n/k \rceil \geq 5$
- This means $n/k > 4$, so $n > 4k$
- Minimum $n = 4k + 1$

If $k = 10$ colors: $n = 4(10) + 1 = \mathbf{41 \text{ people}}$

Problem D2.2: What's the minimum number of students needed in a class to guarantee 3 get the same grade (A, B, C, D, E)?

Solution:

- Want: $\lceil n/5 \rceil \geq 3$

- This means $n/5 > 2$, so $n > 10$
- Minimum $n = \mathbf{11 \text{ students}}$

Level 3: Complex Scenarios

Problem D3.1: A box has 10 red, 8 blue, 12 green, 15 yellow socks. How many to guarantee 6 socks of the same color?

Solution:

- Worst case: Pick 5 of each color = $5 \times 4 = 20$ socks (no guarantee yet)
- The next (21st) sock must match one color, giving 6 socks
- Answer: **21 socks**

Problem D3.2: Among any 25 people, prove that at least 3 share the same birth month.

Solution:

- $n = 25$ (people)
- $m = 12$ (months)
- Minimum = $\lceil 25/12 \rceil = \lceil 2.08 \rceil = 3$

Therefore, among any 25 people, at least **3 must share the same birth month.** ✓

SECTION E: MIXED COMPREHENSIVE PROBLEMS

Problem E1: Let $A = \{1, 2, 3, 4, 5, 6\}$.

Define $R = \{(x,y) \mid x \text{ divides } y \text{ and } x < y\}$.

- a) List all elements of R
- b) Find Domain and Range
- c) Check: Is R reflexive? Symmetric? Transitive?
- d) Construct the relation matrix

Solution:

a) Elements of R:

- 1 divides 2,3,4,5,6: (1,2), (1,3), (1,4), (1,5), (1,6)
- 2 divides 4,6: (2,4), (2,6)
- 3 divides 6: (3,6)
- R =

b) Domain & Range:

- Domain =
- Range =

c) Properties:

- **Reflexive:** No $\times$ (requires $x < y$, so (x,x) can't be in R)
- **Symmetric:** No $\times$ (if $(1,2) \in R$, then $(2,1)$ must be in R, but $2 \nmid 1$)
- **Transitive:** Yes $\checkmark$ (if $x|y$ and $y|z$, then $x|z$, and $x < y < z$)

d) Relation Matrix:

	1	2	3	4	5	6
1	[0	1	1	1	1	1]
2	[0	0	0	1	0	1]
3	[0	0	0	0	0	1]
4	[0	0	0	0	0	0]
5	[0	0	0	0	0	0]
6	[0	0	0	0	0	0]

Problem E2: Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by:

$$f(x) = \begin{cases} 2x - 1 & \text{if } x < 0 \\ x^2 & \text{if } x \geq 0 \end{cases}$$

- a) Find $f(-2)$, $f(0)$, $f(3)$
- b) Find $f^{-1}(3)$
- c) Is f injective?
- d) Is f surjective?

Solution:

a) Function Evaluation:

- $f(-2) = 2(-2) - 1 = -5$ (using first case)
- $f(0) = 0^2 = 0$ (using second case)
- $f(3) = 3^2 = 9$ (using second case)

b) Finding $f^{-1}(3)$:

Need x where $f(x) = 3$

Case 1 ($x < 0$): $2x - 1 = 3 \Rightarrow 2x = 4 \Rightarrow x = 2$, but $2 \geq 0 \times$

Case 2 ($x \geq 0$): $x^2 = 3 \Rightarrow x = \sqrt{3} \geq 0 \checkmark$

$$f^{-1}(3) = \sqrt{3}$$

c) Injective:

Not injective because:

- $f(-2) = -5$
- $f(1) = 1^2 = 1$
- $f(-3) = 2(-3) - 1 = -7$

However, we need to check if different inputs can give same output

- For $x < 0$: $f(x) = 2x - 1$ ranges over $(-\infty, -1)$

- For $x \geq 0$: $f(x) = x^2$ ranges over $[0, \infty)$
- These ranges don't overlap, so f IS injective ✓

d) Surjective:

Range = $(-\infty, -1) \cup [0, \infty) \neq \mathbb{R}$ (missing $[-1, 0)$)

Not surjective ✗

FINAL EXAM-STYLE PROBLEMS

Problem F1 (Proof): Prove that the relation "has the same remainder when divided by 5" on the set of integers is an equivalence relation.

[See Solution: Section B3.1 (modulo 5) - Similar proof structure]

Problem F2 (Bijection Finding): Show that $f: \mathbb{R} \rightarrow (0, \infty)$ defined by $f(x) = e^x$ is bijective and find f^{-1} .

Solution:

Injective:

If $e^a = e^b$, then $a = b$ (exponential is 1-1)

Yes, injective ✓

Surjective:

For any $y \in (0, \infty)$, solve $e^x = y$

$\Rightarrow x = \ln(y) \in \mathbb{R}$

Yes, surjective ✓

Bijjective: Yes ✓

Inverse: $f^{-1}(x) = \ln(x)$ for $x \in (0, \infty)$

Problem F3 (Combined): Let $A = \{1,2,3\}$ and $B = \{a,b\}$.

- a) Find $|A \times B|$
- b) How many relations exist from A to B?
- c) How many functions exist from A to B?
- d) How many are injective?
- e) How many are surjective?
- f) How many are bijective?

Solution:

a) $|A \times B| = 3 \times 2 = \mathbf{6}$

b) Number of relations $= 2^{(|A \times B|)} = 2^6 = \mathbf{64}$

c) Number of functions from A to B:

Each of 3 elements in A chooses 1 of 2 elements in B

$= 2^3 = \mathbf{8 \text{ functions}}$

d) Injective Functions:

If $|A| = 3$ and $|B| = 2$, we can't have injective function ($3 > 2$)

$= \mathbf{0 \text{ injective functions}}$

e) Surjective Functions:

Need to hit both a and b

Using inclusion-exclusion:

- Total functions $= 8$
- Functions missing a $= 1$ (all map to b) $= 1$
- Functions missing b $= 1$ (all map to a) $= 1$
- Surjective $= 8 - 1 - 1 = \mathbf{6 \text{ functions}}$

f) Bijective Functions:

Can't be bijective ($|A| \neq |B|$)

= 0 bijective functions

ANSWER KEY SUMMARY

Section	Problem	Answer
A	A1.1	6 elements
A	A2.2	$X = \{a, b\}$, $Y =$
A	A2.3	3
B	B1.2	$D = \{2, 3, 4\}$, $R =$
B	B2.1	Equivalence relation
C	C1.1	Yes
C	C2.1	Bijjective; $f^{-1}(x) = (x-1)/2$
C	C3.1	$(f \circ g) = 3x+2$, $(g \circ f) = 3x+6$
D	D1.1	2 people
D	D2.2	11 students

Practice Strategy:

1. Solve Level 1 problems without notes
2. Check solutions
3. Attempt Level 2 problems
4. Review Level 3 for deeper understanding
5. Complete mixed and exam-style problems under timed conditions